Dynamic Modeling of Indoor Blimp with Differential Thrust Vectoring

Technical Report

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Contents

1. Introduction

This report presents a comprehensive dynamic model for an indoor blimp equipped with differential thrust vectoring capability. The system utilizes two independently controlled motors with servo-actuated gimbal mechanisms to achieve six degrees of freedom (6-DOF) control in a constrained indoor environment.

1.1. System Overview

The indoor blimp is designed for:

- Low-speed maneuvering in confined spaces
- Stable hovering with precise position control
- Gentle, predictable dynamics suitable for indoor operation
- Energy-efficient operation through buoyancy compensation

Key characteristics:

- Near-neutral buoyancy operation
- Differential thrust for yaw control
- Thrust vectoring for pitch and translation control
- Low Reynolds number aerodynamics
- Minimal environmental disturbances

2. Physical Model

2.1. Coordinate Systems

2.1.1. Body-Fixed Frame

The body-fixed coordinate system $\{\boldsymbol{F}_B\}$ is defined with origin at the center of buoyancy (COB):

- $\boldsymbol{x}_B$: Forward (longitudinal axis)
- $\boldsymbol{y}_B$: Right (lateral axis)
- $\boldsymbol{z}_B$: Down (vertical axis, right-hand rule)

2.1.2. Inertial Frame

The inertial frame $\{\boldsymbol{F}_I\}$ is defined as:

- $\boldsymbol{X}_I$: Fixed horizontal direction
- $\boldsymbol{Y}_I$: Fixed horizontal direction (perpendicular)
- $\boldsymbol{Z}_I$: Down (gravity direction)

2.2. State Vector

The complete state of the blimp is described by:

$$\boldsymbol{x} = \begin{pmatrix} \boldsymbol{p} \\ \boldsymbol{v} \\ \boldsymbol{\Theta} \\ \boldsymbol{\omega} \end{pmatrix} = \begin{pmatrix} x & y & z & u & v & w & \varphi & \theta & \psi & p & q & r \end{pmatrix}^T$$

where:

- $\boldsymbol{p} = [x, y, z]^T$: Position in inertial frame (m)
- $\boldsymbol{v} = [u, v, w]^T$: Linear velocity in body frame (m/s)

- $\mathbf{\Theta} = [\varphi, \theta, \psi]^T$: Euler angles (roll, pitch, yaw) (rad)
- $\boldsymbol{\omega} = [p, q, r]^T$: Angular velocity in body frame (rad/s)

2.3. Physical Parameters

Parameter	**Symbol**	**Typical Value**
Envelope volume	V	0.5 m^3
Total mass	m	0.15 kg
Air density	ρ	1.225 kg/m^3
Buoyant force	F_B	6.0 N
Weight	$W = mg$	1.47 N
Net buoyancy	ΔF	+4.53 N (5% positive)
Envelope length	L	1.2 m
Envelope diameter	D	0.4 m
Motor spacing	d	0.3 m
Max thrust per motor	$T_{\max}$	3.0 N

2.3.1. Inertia Properties

The blimp's moment of inertia tensor in body frame:

$$\boldsymbol{I} = \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{pmatrix}$$

For an ellipsoidal envelope with uniform mass distribution:

$$I_{xx} = \frac{m}{5}(b^2 + c^2) \approx 0.008 \text{kg}\cdot\text{m}^2$$

$$I_{yy} = \frac{m}{5}(a^2 + c^2) \approx 0.032 \text{kg}\cdot\text{m}^2$$

$$I_{zz} = \frac{m}{5}(a^2 + b^2) \approx 0.032 \text{kg}\cdot\text{m}^2$$

where $a = \frac{L}{2}$, $b = c = \frac{D}{2}$ are semi-axes.

3. Equations of Motion

3.1. Translational Dynamics

The translational equations of motion in body frame:

$$m(\dot{\boldsymbol{v}} + \boldsymbol{\omega} \times \boldsymbol{v}) = \boldsymbol{F}_{\text{thrust}} + \boldsymbol{F}_{\text{aero}} + \boldsymbol{R}^T \boldsymbol{F}_{\text{grav}}$$

Expanded component form:

$$\begin{aligned}
m(\dot{u} - vr + wq) &= F_x + F_{\text{aero,x}} - mg\sin(\theta) \\
m(\dot{v} - wp + ur) &= F_y + F_{\text{aero,y}} + mg\cos(\theta)\sin(\varphi) \\
m(\dot{w} - uq + vp) &= F_z + F_{\text{aero,z}} + mg\cos(\theta)\cos(\varphi) - F_B
\end{aligned}$$

where $\boldsymbol{R}$ is the rotation matrix from inertial to body frame.

3.2. Rotational Dynamics

Euler's rotation equations:

$$\boldsymbol{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\boldsymbol{I}\boldsymbol{\omega}) = \boldsymbol{M}_{\text{thrust}} + \boldsymbol{M}_{\text{aero}}$$

Component form:

$$\begin{aligned}
I_{xx}\dot{p} + \left(I_{zz} - I_{yy}\right)qr &= M_x \\
I_{yy}\dot{q} + (I_{xx} - I_{zz})pr &= M_y \\
I_{zz}\dot{r} + \left(I_{yy} - I_{xx}\right)pq &= M_z
\end{aligned}$$

3.3. Kinematic Relations

Position rate:

$$\dot{\boldsymbol{p}} = \boldsymbol{R}\boldsymbol{v}$$

Euler angle rates:

$$\begin{pmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} 1 & \sin(\varphi)\tan(\theta) & \cos(\varphi)\tan(\theta) \\ 0 & \cos(\varphi) & -\sin(\varphi) \\ 0 & \sin(\varphi)\sec(\theta) & \cos(\varphi)\sec(\theta) \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

4. Thrust Model

4.1. Motor Configuration

The blimp uses two motors symmetrically placed:

- Motor 1: Position $\boldsymbol{r}_1 = \left[-\frac{d}{2}, 0, 0\right]^T$ (left)
- Motor 2: Position $\boldsymbol{r}_2 = \left[+\frac{d}{2}, 0, 0\right]^T$ (right)

Each motor can gimbal in the x-z plane with angle δ_i.

4.2. Thrust Vector Calculation

For motor i with gimbal angle δ_i and thrust magnitude T_i:

$$\boldsymbol{T}_i = T_i \begin{pmatrix} \sin(\delta_i) \\ 0 \\ -\cos(\delta_i) \end{pmatrix}$$

where:
- $\delta_i = 0$: Vertical thrust (hover)
- $\delta_i > 0$: Thrust tilted forward
- $\delta_i < 0$: Thrust tilted backward

Typical range: $\delta_i \in [-30°, +30°]$

4.3. Total Forces and Moments

Total thrust force in body frame:

$$\boldsymbol{F}_{\text{thrust}} = \boldsymbol{T}_1 + \boldsymbol{T}_2 = \begin{pmatrix} T_1 \sin(\delta_1) + T_2 \sin(\delta_2) \\ 0 \\ -(T_1 \cos(\delta_1) + T_2 \cos(\delta_2)) \end{pmatrix}$$

Total moment about center of buoyancy:

$$\boldsymbol{M}_{\text{thrust}} = \boldsymbol{r}_1 \times \boldsymbol{T}_1 + \boldsymbol{r}_2 \times \boldsymbol{T}_2$$

Expanded:

$$\begin{aligned} M_x &= 0 \quad \text{(no roll moment from symmetric thrust)} \\ M_y &= -\frac{d}{2}(T_1 \sin(\delta_1) + T_2 \sin(\delta_2)) \quad \text{(pitch moment)} \\ M_z &= \frac{d}{2}(T_1 \cos(\delta_1) - T_2 \cos(\delta_2)) \quad \text{(yaw moment)} \end{aligned}$$

5. Aerodynamic Model

5.1. Drag Forces

Due to low speeds (typically < 2 m/s) and low Reynolds number ($\text{Re} \approx 10^4$), aerodynamic forces are dominated by viscous drag:

$$\boldsymbol{F}_{\text{aero}} = -\begin{pmatrix} K_u u \ |u| \\ K_v v \ |v| \\ K_w w \ |w| \end{pmatrix}$$

Drag coefficients (empirical):
- $K_u \approx 0.5 \ \text{N} \cdot \text{s}^2/\text{m}^2$ (longitudinal)
- $K_v \approx 1.5 \ \text{N} \cdot \text{s}^2/\text{m}^2$ (lateral - higher due to envelope profile)
- $K_w \approx 1.2 \ \text{N} \cdot \text{s}^2/\text{m}^2$ (vertical)

5.2. Aerodynamic Moments

Damping moments resist angular motion:

$$\boldsymbol{M}_{\text{aero}} = -\begin{pmatrix} K_p p \ |p| \\ K_q q \ |q| \\ K_r r \ |r| \end{pmatrix}$$

Moment damping coefficients:

- $K_p \approx 0.005 \text{ N} \cdot \text{m} \cdot \text{s}^2$
- $K_q \approx 0.012 \text{ N} \cdot \text{m} \cdot \text{s}^2$
- $K_r \approx 0.012 \text{ N} \cdot \text{m} \cdot \text{s}^2$

5.3. Added Mass Effects

For a body moving through fluid, added mass must be considered:

$$(m + m_a)\dot{\boldsymbol{v}} = \boldsymbol{F}$$

Added mass coefficients for ellipsoid:

- $m_{a,x} \approx 0.1m$ (longitudinal)
- $m_{a,y} \approx 0.5m$ (lateral)
- $m_{a,z} \approx 0.5m$ (vertical)

6. Control Strategy

6.1. Control Inputs

The system has 4 control inputs:

$$\boldsymbol{u}_{\text{control}} = \begin{pmatrix} \delta_1 \\ T_1 \\ \delta_2 \\ T_2 \end{pmatrix}$$

Subject to constraints:

$$\delta_i \in [-30°, +30°]$$
$$T_i \in [0, T_{\max}]$$

6.2. Decoupled Control Modes

6.2.1. Vertical Control (Altitude)

Average vertical thrust:

$$F_z = -(T_1 \cos(\delta_1) + T_2 \cos(\delta_2))$$

For hover: $F_z \approx W - F_B$

6.2.2. Longitudinal Control (Forward/Backward)

Average horizontal thrust:

$$F_x = T_1 \sin(\delta_1) + T_2 \sin(\delta_2)$$

Also creates pitch moment for attitude control.

6.2.3. Yaw Control (Rotation)

Differential vertical thrust:

$$M_z = \frac{d}{2}(T_1 \cos(\delta_1) - T_2 \cos(\delta_2))$$

For pure yaw: $T_1 \neq T_2$ with $\delta_1 = \delta_2 = 0$

6.2.4. Pitch Control (Nose Up/Down)

Pitch moment from thrust vectoring:

$$M_y = -\frac{d}{2}(T_1 \sin(\delta_1) + T_2 \sin(\delta_2))$$

6.3. Control Allocation

For desired forces and moments (F_x^*, F_z^*, M_z^*):

$$\begin{pmatrix} \sin(\delta_1) & \sin(\delta_2) \\ -\cos(\delta_1) & -\cos(\delta_2) \\ \frac{d}{2}\cos(\delta_1) & -\frac{d}{2}\cos(\delta_2) \end{pmatrix} \begin{pmatrix} T_1 \\ T_2 \end{pmatrix} = \begin{pmatrix} F_x^* \\ F_z^* \\ M_z^* \end{pmatrix}$$

This is an overdetermined system solved via pseudoinverse or optimization.

7. Stability Analysis

7.1. Equilibrium Points

For steady hover with zero velocity:

$$\begin{aligned} \boldsymbol{v} &= \mathbf{0} \\ \boldsymbol{\omega} &= \mathbf{0} \\ \theta &= 0, \quad \varphi = 0 \end{aligned}$$

Thrust requirement:

$$T_1 + T_2 = \frac{W - F_B}{\cos(\delta)}$$

7.2. Linearized Dynamics

Small perturbation analysis about hover yields decoupled subsystems:

Longitudinal (x-z plane):

$$\begin{pmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} -\frac{K_u}{m} & 0 & 0 & -g \\ 0 & -\frac{K_w}{m} & 0 & 0 \\ 0 & 0 & -\frac{K_q}{I_y} & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} u \\ w \\ q \\ \theta \end{pmatrix} + \begin{pmatrix} \frac{1}{m} \\ 0 \\ -\frac{d}{2I_y} \\ 0 \end{pmatrix} F_x$$

Lateral-Directional (y-ψ plane):

$$\begin{pmatrix} \dot{v} \\ \dot{r} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} -\frac{K_v}{m} & 0 & 0 \\ 0 & -\frac{K_r}{I_z} & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} v \\ r \\ \psi \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{d}{2I_z} \\ 0 \end{pmatrix} M_z$$

Both subsystems are stable due to aerodynamic damping.

8. Simulation Results

8.1. Numerical Integration

The equations of motion are integrated using 4th-order Runge-Kutta method with time step $\Delta t = 0.01$ s.

8.2. Maneuver Examples

8.2.1. Hover Stability

- Initial: $\boldsymbol{v} = \mathbf{0}$, $z = 1$ m
- Control: $T_1 = T_2 = \frac{W - F_B}{2}$, $\delta_1 = \delta_2 = 0$
- Result: Stable hover with < 5 cm drift over 60 seconds

8.2.2. Forward Flight

- Command: $F_x = 0.5$ N
- Control: $\delta_1 = \delta_2 = 15°$, adjust thrust for altitude
- Result: Smooth acceleration to 0.8 m/s cruise speed

8.2.3. Yaw Turn

- Command: $M_z = 0.02$ N $\cdot$ m
- Control: $T_1 = 1.2T_{\text{eq}}$, $T_2 = 0.8T_{\text{eq}}$, $\delta = 0$
- Result: 45° turn in 3 seconds with minimal altitude change

9. Design Considerations

9.1. Buoyancy Selection

Net buoyancy affects control authority and efficiency:

Condition	**Advantages**	**Disadvantages**
Positive (5-10%)	Safety, easier hover	Constant downward thrust needed
Neutral (±2%)	Energy efficient, agile	Difficult to maintain, unstable
Negative	Stable in floor effect	Cannot hover, limited altitude

Recommendation: +5% positive buoyancy for indoor safety and ease of control.

9.2. Motor Placement

Optimal motor spacing d balances:

- Yaw authority: $M_z \propto d$
- Structural weight: increases with d
- Stability margin: wider spacing improves

Recommendation: $d = 0.25L$ for good control without excessive weight.

9.3. Gimbal Range

Servo angle limits affect maneuverability:

- $\pm 15°$: Gentle maneuvers, energy efficient
- $\pm 30°$: Aggressive control, full 6-DOF capability
- $\pm 45°$: Extreme maneuvers, reduced vertical efficiency

Recommendation: $\pm 25°$ provides good balance for indoor operation.

10. Conclusion

The presented dynamic model captures the essential physics of an indoor blimp with differential thrust vectoring:

1. **6-DOF Control:** Achievable with only two motors through combined differential thrust and vectoring
2. **Inherent Stability:** Aerodynamic damping and buoyancy provide stable, predictable dynamics
3. **Energy Efficiency:** Buoyancy compensation reduces power requirements significantly
4. **Indoor Suitability:** Gentle dynamics and large envelope provide safe indoor operation

10.1. Future Work

- Experimental validation of aerodynamic coefficients
- Adaptive control for varying payload and envelope pressure
- Vision-based state estimation for indoor navigation
- Obstacle avoidance integration
- Multi-blimp coordination strategies

11. References

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2. Mueller, J. B. et al. "Development of an Indoor Micro Blimp," *AIAA Guidance, Navigation, and Control Conference*, 2007.
3. Hygounenc, E. et al. "The Autonomous Blimp Project of LAAS-CNRS," *International Conference on Robotics and Automation*, 2004.
4. Elfes, A. et al. "Autonomous Flight Control for a Titan Exploration Aerobot," *AIAA Space Conference*, 2003.

12. Appendix A: Control Allocation Algorithm

```python
import numpy as np

def allocate_control(F_x_desired, F_z_desired, M_z_desired,
                     d=0.3, T_max=3.0):
    """
    Allocate thrust commands to achieve desired forces and moments.

    Parameters:
        F_x_desired: Desired horizontal force (N)
        F_z_desired: Desired vertical force (N)
        M_z_desired: Desired yaw moment (N·m)
        d: Motor spacing (m)
        T_max: Maximum thrust per motor (N)

    Returns:
        delta_1, T_1, delta_2, T_2: Control commands
    """

    # Start with symmetric vectoring for F_x
    if abs(F_x_desired) > 0.01:
        delta_avg = np.arctan2(F_x_desired, -F_z_desired)
        delta_avg = np.clip(delta_avg, -np.pi/6, np.pi/6)
    else:
        delta_avg = 0.0

    # Calculate required total thrust
    T_total = np.sqrt(F_x_desired**2 + F_z_desired**2)

    # Differential thrust for yaw
    if abs(F_z_desired) > 0.01:
        Delta_T = 2 * M_z_desired / (d * np.cos(delta_avg))
    else:
        Delta_T = 0.0

    # Allocate to individual motors
    T_1 = (T_total + Delta_T) / 2
    T_2 = (T_total - Delta_T) / 2
    delta_1 = delta_2 = delta_avg

    # Saturate thrusts
    T_1 = np.clip(T_1, 0, T_max)
    T_2 = np.clip(T_2, 0, T_max)

    return delta_1, T_1, delta_2, T_2
```

13. Appendix B: Simulation Parameters

Complete parameter set for numerical simulation:

Category	Parameter	Value

Mass Properties	Total mass m	0.15 kg
	I_{xx}	$0.008\ \mathrm{kg \cdot m^2}$
	I_{yy}	$0.032\ \mathrm{kg \cdot m^2}$
	I_{zz}	$0.032\ \mathrm{kg \cdot m^2}$
Geometry	Length L	1.2 m
	Diameter D	0.4 m
	Motor spacing d	0.3 m
Aerodynamics	K_u	$0.5\ \mathrm{N \cdot s^2/m^2}$
	K_v	$1.5\ \mathrm{N \cdot s^2/m^2}$
	K_w	$1.2\ \mathrm{N \cdot s^2/m^2}$
	K_p	$0.005\ \mathrm{N \cdot m \cdot s^2}$
	K_q	$0.012\ \mathrm{N \cdot m \cdot s^2}$
	K_r	$0.012\ \mathrm{N \cdot m \cdot s^2}$
Buoyancy	Volume V	$0.5\ \mathrm{m^3}$
	Buoyant force F_B	6.0 N
	Net buoyancy	+4.53 N
Actuation	Max thrust $T_{\max}$	3.0 N
	Gimbal range	±30°
	Servo bandwidth	10 Hz